

PRATHAMESH SARAF

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EDUCATION

University of California, San Diego

San Diego, USA

MS in Electrical & Computer Engineering (Intelligent Systems, Robotics, & Controls)

2022 - 2024

Courses: Robot Motion Planning, Sensing and Estimation, Co-operative Control of Multi-Agent Systems, Statistical Learning, Convex Optimization, Linear Algebra, Non-Linear Controls, Stochastic Processes in Dynamic Systems

Birla Institute of Technology & Science, Pilani

Hyderabad, India

B.E. Electronics & Instrumentation; Minor in Robotics & Automation

2017-2021

Courses: Modern Control Systems, Control Systems, AI for Robotics, Robotics

SKILLS

Programming & Scripting: Python, C/C++, RUST, Java, HTML, Verilog, PLC Programming Simulation

Robotics & Control Tools: MATLAB, ROS, Simulink, Gazebo, Webots, PyBullet, MuJoCo, LabVIEW

CAD/Hardware: SolidWorks, SolidWorks Electrical, Fusion 360, AutoCAD, DraftSight, HDL, Allen-Bradley PLC, Test bench

EXPERIENCE

Controls Engineer, ASML

June 2024 - Present

- Designed a closed-loop control for the Energy Controller using system identification by modeling the plant through Bode, Nyquist, and step response analyses; computed optimal gains and validated performance via simulation and source testing. Also implemented feedforward control to improve behavior in edge cases.
- Led end-to-end delivery of open-loop control: updated requirements and the Simulink model, designed test cases, performed unit/bench testing, and coordinated with cross-functional team integration to deliver ahead of schedule.
- Built the 'Sensor Acquisition' HIL simulator using Simulink-HDL, deployed it on an FPGA, and validated it with regression tests on a testbench. Revived and upgraded the Python simulator to emulate the hardware for High-NA machines.
- Provided rapid escalation and integration support by debugging issues in less than 1 day, delivering FW/control patches, and assisting proto engineers during feature bring-up.

Research Intern, Legged Robot Control & Locomotion Internships

Jan 2021 - Sept 2023

- Developed and optimized gait transition, locomotion, and whole-body control strategies for bipedal, quadruped, and hexapod robots across ROS, PyBullet, Gazebo, and MuJoCo platforms, improving stability and agility on uneven terrains.
- Implemented motion imitation, Model Predictive Control, and Central Pattern Generator-based methods with real-time simulation tools; led a 4-member team and collaborated with major research institutes like IISc and NUS

Electronics Subsystem Lead, Hyperloop India

Aug 2019 - Oct 2020

- Developed a real-time Kalman filter for Hyperloop pod pose estimation, reducing acceleration RMSE by **91.7%** and velocity RMSE by **67.4%**, critical for precise vehicle control and motion planning.
- Designed an on-board electronics vehicle controller architecture using CAN bus, with localized control loops for motor fine-tuning and executing an Emergency Braking Protocol for enhanced pod safety and operational reliability up to 98%.
- Engineered a comprehensive sensor fusion system, integrating IMUs, tachometers, optical encoders, and fiducial markers for accurate state estimation and feedback, validated through simulations over 120 discrete time-steps.

RELEVANT PROJECTS

Traffic Wave Dampening using Autonomous Vehicles | Python

Apr 2023 - Jun 2023

- Modeled complex traffic flow dynamics using a state-space representation to validate Lagrangian control strategies for autonomous vehicles, proving a single AV's capacity to significantly dampen traffic waves and ensure system stabilizability.
- Implemented a Follower Stopper controller, reducing velocity standard deviation by **80.8%**, fuel consumption by **42.5%**, and excessive braking by **98.6%**, while concurrently increasing traffic throughput by **14.1%**.

Multi-Modal Sensor Fusion for Robotic Localization and Mapping | Python

Jan 2023 - Mar 2023

- Implemented comprehensive Simultaneous Localization and Mapping solutions for autonomous robots, fusing IMUs, encoders, 2D LiDAR (up to 30m range), and stereo/RGBD camera data to construct 2D occupancy grids to track landmarks
- Optimized a 100-particle Particle Filter for differential-drive robot localization and an Extended Kalman Filter (EKF) for visual-inertial SLAM, ensuring robust and accurate pose tracking.
- Validated high-fidelity sensor processing pipelines, encompassing IMU calibration, 10Hz yaw rate filtering, and gradient descent optimization for quaternion-based orientation tracking for precise vehicle control and panoramic mapping.

A Convolutional Neural Network Approach Towards Self-Driving Cars | C++, Python

Sept 2018 - Mar 2019

- Implemented a Convolutional Neural Network (CNN) for Level 2 autonomous vehicle steering control, achieving a Mean Squared Error (MSE) loss of **0.0003798** after 30 epochs by directly mapping camera input to steering commands.
- Integrated a robust obstacle avoidance system with ultrasonic sensors and the RRT*-Connect algorithm, enabling autonomous lane changes based on obstacle velocity tracking within a **20cm** detection threshold.
- Engineered the vehicle's control architecture, integrating a Raspberry Pi for image processing and an Arduino Mega for motor control, resulting in an **86.43%** autonomy rate during testing on the CARLA open-source driving simulator while adhering to ISO26262 functional safety principles.

OTHER PUBLICATIONS / PROJECT REPORTS

- P. Saraf**, M. Shaikh, M. Phan, "[Convex Optimization in Legged Robots](#)," Project Report MAE 227
- P. Saraf**, Y. Jangir, R. N. Ponnalagu, "[Implementation and Testing of Force Control on a Spherical Articulated Manipulator](#)," IEEE International Conference on Mechatronics and Automation (ICMA), 2022
- P. Saraf**, et al, "[Terrain Adaptive Gait Transitioning for a Quadruped Robot using Model Predictive Control](#)," IEEE International Conference on Automation and Control (ICAC), 2021
- P. Saraf**, R. N. Ponnalagu, "[Modeling and Simulation of a Point-to-Point Spherical Articulated Manipulator Using Optimal Control](#)," IEEE International Conference on Automation, Robotics, and Applications (ICARA), 2021
- P. Saraf**, M. Gupta, and A.M. Parimi, "[A Comparative Study Between a Classical and Optimal Controller for a Quadrotor](#)," IEEE 17th India Council International Conference (INDICON), 2020

TEACHING / LEADERSHIP

- Teaching Assistant** - Control Systems Laboratory, BITS Pilani
- Robotics Mentor** - Student Mentorship Program, BITS Pilani

Jan 2020 - May 2020

Aug 2019 - Dec 2019